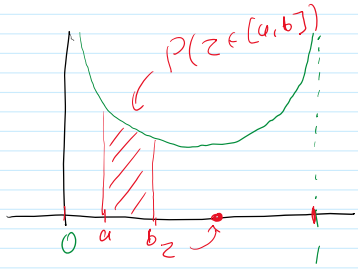
14. Review and Bootstrap

February 6, 2026 9:02 AM

We write $Z \sim \text{Beta}(1/2, 1/2)$

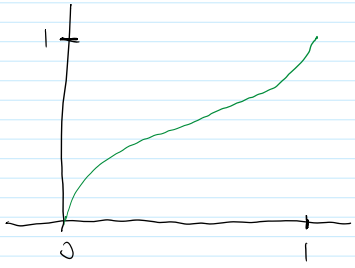
$$h(z) = \frac{1}{\pi} \frac{1}{\sqrt{z(1-z)}}$$



$$P(Z \in [a,b]) = \int_a^b h(z)\,dz$$

$$AR = \frac{f(x)}{h(x)}$$

PIT:



CDF: $H(x) = \arcsin(x)$

$= P(Z \le x)$

$= \int_0^x h(t)\,dt$

AR: • Say $f(x)$ has a domain / support on $[-1, 1]$

i.e. $f(x) > 0$ on $[-1, 1]$

• take $h(x) = \frac{1}{4} \mathbb{1}[-2, 2]$ ← unif$[-2, 2]$

$$\frac{f(x)}{h(x)} = \frac{\frac{3}{4}(1-x^2)\mathbb{1}[-1,1]}{1/4} = \underset{c=3}{3}\,\underbrace{(1-x^2)}_{g(x) \in [0,1] \text{ for } x \in [-1,1]} = c \cdot g(x)$$

$$P(\text{Accept}) = \frac{1}{c} = \frac{1}{3}$$

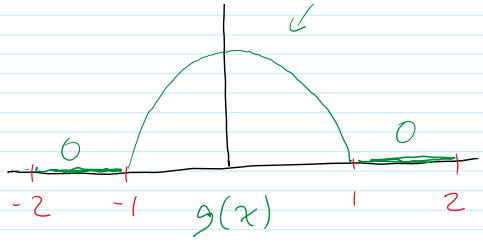
Algorithm:

- Sample $Z \sim \text{Unif}[-2, 2]$
- Sample $U \sim \text{Unif}[0, 1]$
- Compute $g(Z) = (1 - Z^2) \times \mathbb{1}[Z \in [-1, 1]]$
- If $U \le g(Z)$
  Accept

Accept

Else

Reject



$Z \sim Unif[-2, 2]$,

with Probability $1/2$ we pick a point in $[-2,-1] \cup [1,2]$

and $g(Z) = 0$

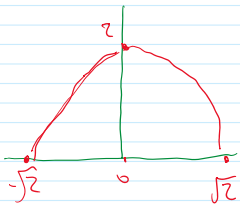
$$P(U \leq 0) = 0$$

50% of the time we Automatically reject as we picked a Z outside of the domain $[-1,1]$

Lets say that $\frac{f(x)}{h(x)} = 3(2 - x^2)$

$$c = \max_{[-2,2]} \frac{f(x)}{h(x)} = \max_{[-2,2]} \{3(2 - x^2)\}$$

$$= 3 \cdot 2 = 6$$



Plugging in x=0:
3(2 - 0^2) = 3(2) = 6.

$$g(x) = \frac{f(x)}{c\,h(x)} = \frac{1}{2}(2 - x^2)$$

Jackknife: Computed n LOO estimates based on taking away one data point at a time (No Randomization)

(1950's)

Bootstrap: "resampling Bootstrap"

(1979)

- resample my dataset with replacement
- Do this many times
- Recompute the estimate $\hat{\theta}$ each time.

(Randomized)

Dataset $\{X_1, \dots, X_n\}$

- $\{\underbrace{X_1, X_1, X_1}_{\text{show up More than once}}, X_3, X_3, X_2, \dots\}$ ← resample n data points with replacement

X_2 don't occur

For $B \in \mathbb{N}$

More than once

x_2 won't occur

For $B \in \mathbb{N}$

For each Bootstrap data set, compute $\hat{\theta}_1, \ldots, \hat{\theta}_B$

e.g. Estimate the variance: $\frac{1}{B-1} \sum_{i=1}^{B} \left(\hat{\theta}_i - \bar{\hat{\theta}} \right)^2$

Sample Mean of the BS estimates.